HDOC DAY 64

PROBLEM

Write a C program that reads the size and elements of a variable-length integer array, nums[]. The array contains exactly one element repeated once. Pass the array to a function named duplicate_element() using its address. Use pointers to pass the array between the calling function and duplicate_element(). The function duplicate_element() must:

- Find the only repeated element in nums[] and display the repeated element.
- Find the result in a single iteration through the array by using an efficient approach such as the XOR technique.
- Assume that every other element appears exactly once.

CODE

```
#include <stdio.h>


void duplicate_element(int *nums, int n)
{
   int duplicate = 0;


   for (int i = 0; i < n; i++)
   {
      duplicate ^= nums[i];


      if (i < n - 1)
         duplicate ^= (i + 1);
   }
```

```
    printf("%d", duplicate);
}

int main()
{
    int n;

    scanf("%d", &n);

    int nums[n];

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &nums[i]);
    }

    duplicate_element(nums, n);

    return 0;
}
```